

HDOC – 36th Day

Name – Piyush Negi

Sap_id – 590042185

Problem Statement:

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

Algorithm:

1. Start the program.
2. Declare variables choice and character array string.
3. Display the menu:
 - Count characters
 - Print each character
 - Count vowels and consonants
 - Exit
4. Read the user's choice.
5. Use a switch statement to perform the selected operation.
6. If choice = 1:

- Input the string.
- Call countCharacters().
- Initialize count = 0.
- Traverse the string until '\0' is reached.
- Increment count for every character.
- Display the total number of characters.

7. If choice = 2:

- Input the string.
- Call printCharacters().
- Traverse the string until '\0'.
- Print each character on a new line.

8. If choice = 3:

- Input the string.
- Call countVowelsConsonants().
- Initialize vowels = 0 and consonants = 0.
- Traverse each character of the string.
- If the character is a, e, i, o, u (uppercase or lowercase), increment vowels.
- Otherwise, increment consonants.
- Display the number of vowels and consonants.

9. If choice = 4, exit the program.

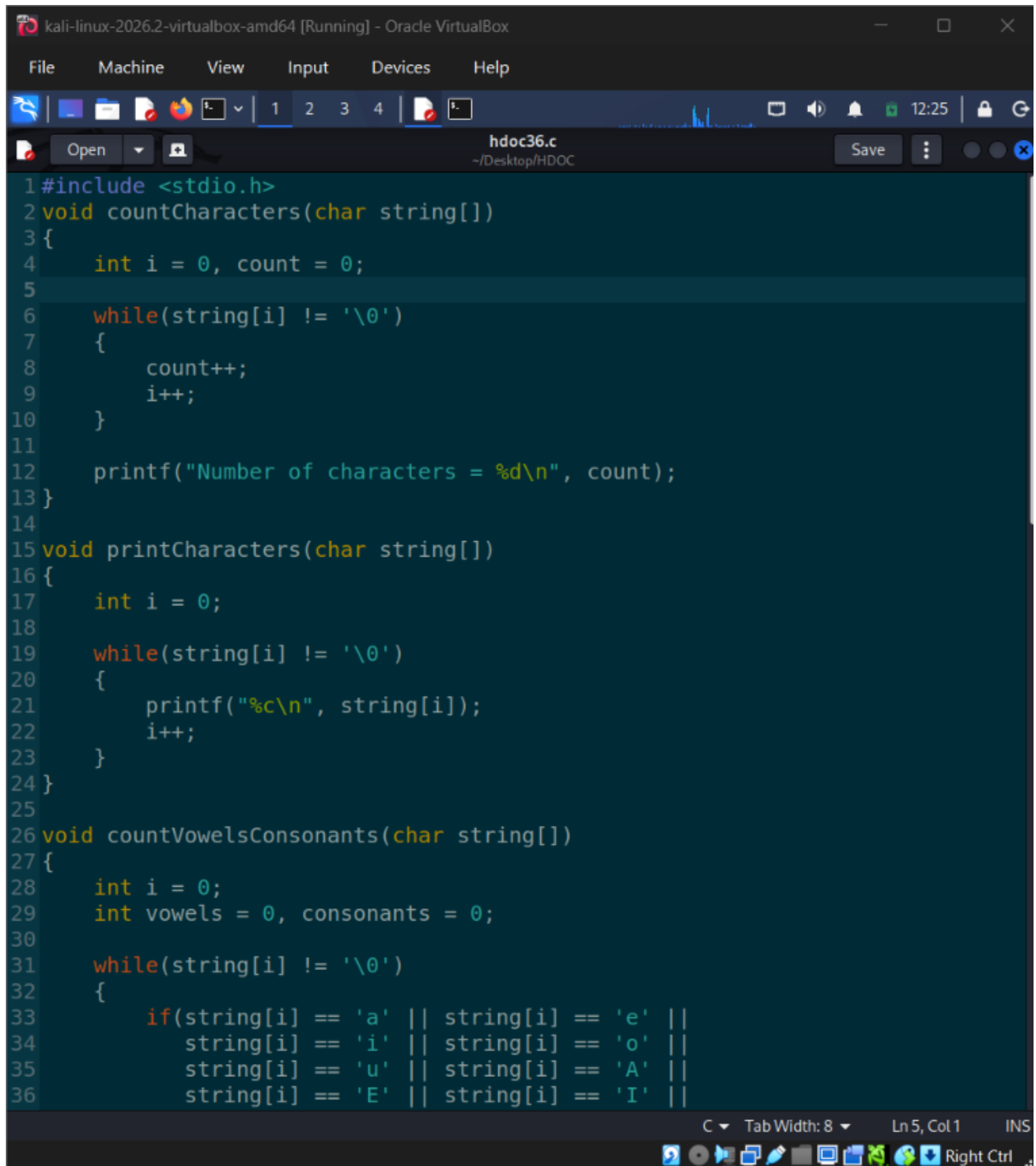
10. For any other choice, display "Invalid choice".

11. Stop.

Test Case Table:

Test Case	Choice	Input	Expected output	Actual output	Result
1	1	Piyush	6	6	Pass
2	2	Piyush	P I Y U S H	P I Y U S H	Pass
3	3	Piyush	Vowels = 2 Consonants = 4	Vowels = 2 Consonants = 4	Pass
4	4	_____	Program exited	Program exited	Pass
5	5	_____	Invalid choice	Invalid choice	Pass

C Program:

A screenshot of a code editor window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The editor shows a C program named "hdoc36.c" located at "~/Desktop/HDOC". The code defines three functions: "countCharacters", "printCharacters", and "countVowelsConsonants". The "countCharacters" function uses a while loop to count the number of characters in a string. The "printCharacters" function uses a while loop to print each character of a string on a new line. The "countVowelsConsonants" function uses a while loop and an if statement to count vowels and consonants in a string. The code is as follows:

```
1#include <stdio.h>
2void countCharacters(char string[])
3{
4    int i = 0, count = 0;
5
6    while(string[i] != '\0')
7    {
8        count++;
9        i++;
10    }
11
12    printf("Number of characters = %d\n", count);
13}
14
15void printCharacters(char string[])
16{
17    int i = 0;
18
19    while(string[i] != '\0')
20    {
21        printf("%c\n", string[i]);
22        i++;
23    }
24}
25
26void countVowelsConsonants(char string[])
27{
28    int i = 0;
29    int vowels = 0, consonants = 0;
30
31    while(string[i] != '\0')
32    {
33        if(string[i] == 'a' || string[i] == 'e' ||
34           string[i] == 'i' || string[i] == 'o' ||
35           string[i] == 'u' || string[i] == 'A' ||
36           string[i] == 'E' || string[i] == 'I' ||
```

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

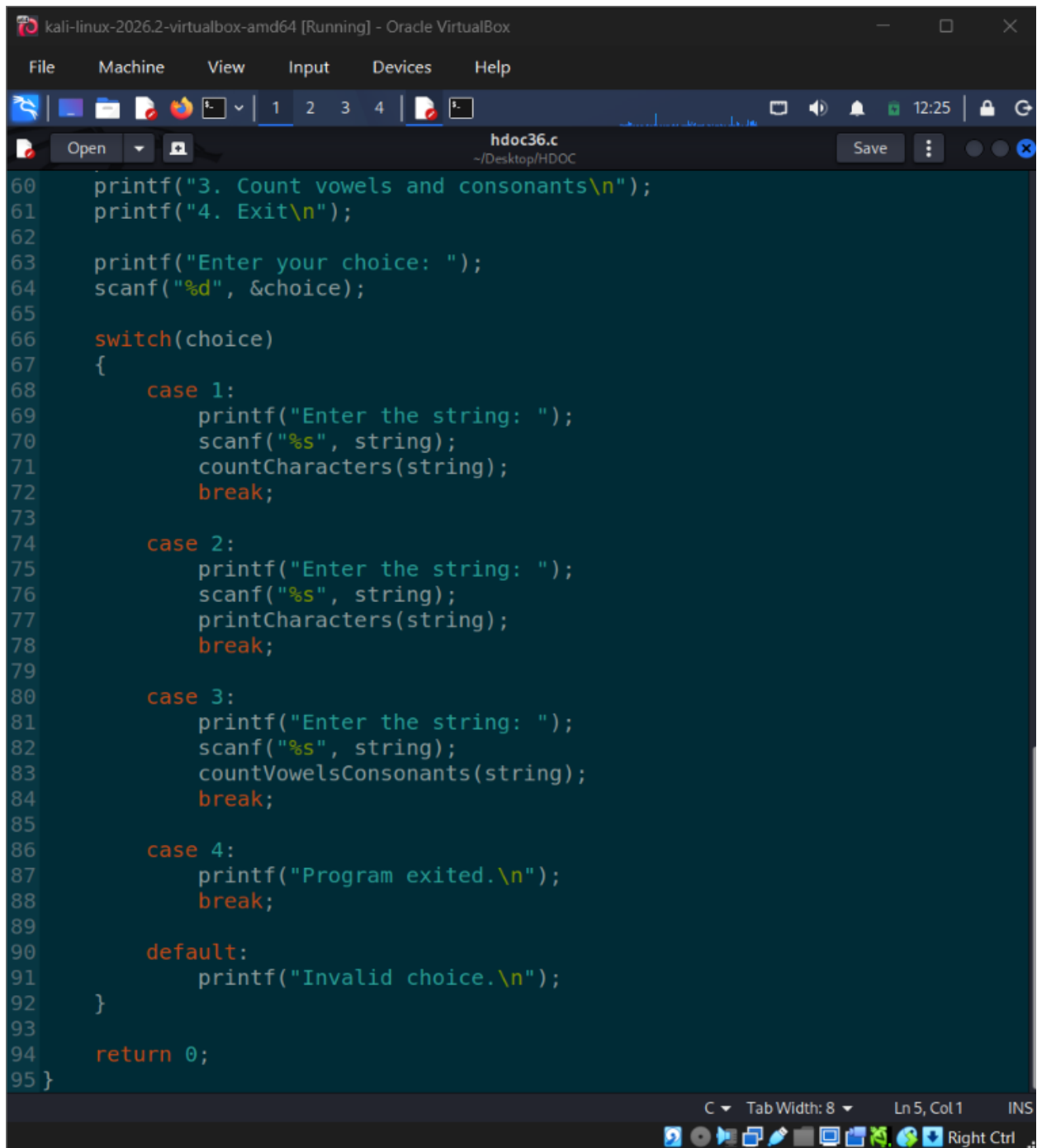
hdoc36.c
~/Desktop/HDOC

Save

```
35     string[i] == 'u' || string[i] == 'A' ||
36     string[i] == 'E' || string[i] == 'I' ||
37     string[i] == 'O' || string[i] == 'U')
38     {
39         vowels++;
40     }
41     else
42     {
43         consonants++;
44     }
45
46     i++;
47 }
48
49 printf("Number of vowels = %d\n", vowels);
50 printf("Number of consonants = %d\n", consonants);
51 }
52
53 int main()
54 {
55     int choice;
56     char string[100];
57
58     printf("1. Count characters\n");
59     printf("2. Print each character\n");
60     printf("3. Count vowels and consonants\n");
61     printf("4. Exit\n");
62
63     printf("Enter your choice: ");
64     scanf("%d", &choice);
65
66     switch(choice)
67     {
68         case 1:
69             printf("Enter the string: ");
70             scanf("%s", string);
```

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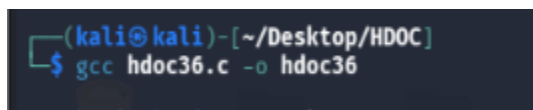
Right Ctrl



```
60 printf("3. Count vowels and consonants\n");
61 printf("4. Exit\n");
62
63 printf("Enter your choice: ");
64 scanf("%d", &choice);
65
66 switch(choice)
67 {
68     case 1:
69         printf("Enter the string: ");
70         scanf("%s", string);
71         countCharacters(string);
72         break;
73
74     case 2:
75         printf("Enter the string: ");
76         scanf("%s", string);
77         printCharacters(string);
78         break;
79
80     case 3:
81         printf("Enter the string: ");
82         scanf("%s", string);
83         countVowelsConsonants(string);
84         break;
85
86     case 4:
87         printf("Program exited.\n");
88         break;
89
90     default:
91         printf("Invalid choice.\n");
92 }
93
94 return 0;
95 }
```

Compile the program using:

`gcc hdoc36.c -o hdoc36`



```
(kali㉿kali)-[~/Desktop/HDOC]
$ gcc hdoc36.c -o hdoc36
```

Execute the program using:

./hdoc36

```
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc36.c -o hdoc36

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc36
1. Count characters
2. Print each character
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Enter the string: Piyush
Number of characters = 6
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc36
1. Count characters
2. Print each character
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Enter the string: Piyush
P
i
y
u
s
h
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc36
1. Count characters
2. Print each character
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Enter the string: Piyush
Number of vowels = 2
Number of consonants = 4
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc36
1. Count characters
2. Print each character
3. Count vowels and consonants
4. Exit
Enter your choice: 5
Invalid choice.

(kali@kali)-[~/Desktop/HDOC]
$
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc36
1. Count characters
2. Print each character
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Program exited.
```